

Ryan M. Richard — Chemistry

Ames National Laboratory and Iowa State University – Ames, IA – USA

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🌐 <https://rmrresearch.github.io/>

Staff Scientist at Ames National Laboratory and Adjunct Assistant Professor of Chemistry at Iowa State University. Research interests include: *ab initio* methods development, computational chemistry, high-performance computing, high-accuracy quantum chemistry, and scientific software design.

Education

- **Ph.D. The Ohio State University** **Columbus, OH**
Dissertation topic: 2008–2013
“Increasing the Computational Efficiency of Ab Initio Methods with Generalized Many-Body Expansions”
Advisor: John M. Herbert
- **B.S. Cleveland State University** **Cleveland, OH**
Undergraduate: Chemistry major, Advisor: David W. Ball 2004–2008

Professional Experience

- **Scientist II, Ames National Laboratory** **Ames, IA**
Architect and lead developer of NWChemEx. 2018–Present
Additional Responsibilities: Mentor undergraduate researchers.
- **Research Assistant, NASA Glenn Research Center** **Cleveland, OH**
Research Topics: 2007–2008
Characterization of degradation properties of ionic liquid based lubricants for terrestrial and space applications by Raman and infrared spectroscopies as well as liquid and gas chromatography.

Academic Experience

- **Adjunct Assistant Professor of Chemistry, Iowa State University** **Ames, IA**
High-accuracy benchmark development. 2022–Present
Additional Responsibilities: Co-advise graduate students, serve on thesis committees.
- **Postdoctoral Researcher, Ames National Laboratory** **Ames, IA**
Algorithm development, development of massively parallel architecture 2017–2018
Additional Responsibilities: Mentor undergraduate researchers
- **Postdoctoral Researcher, Georgia Institute of Technology** **Atlanta, GA**
Algorithm development, high-accuracy benchmarking 2014–2017
Additional Responsibilities: Mentor undergraduate researchers, substitute teach lectures, assist in grant writing
- **Research Assistant, The Ohio State University** **Columbus, OH**
Algorithm development, excited state modeling 2008–2014
- **Teaching Assistant, The Ohio State University** **Columbus, OH**
General chemistry laboratory and recitation, physical chemistry recitation 2009–2011
Additional Responsibilities: Grading, weekly office hours
- **Research Assistant, Cleveland State University** **Cleveland, OH**
Characterization of new high-energy materials 2006–2008

Software Development Experience

- **NWChemEx** <https://github.com/NWChemEx>
2017–present
 - *Architect and lead developer*
 - Package focused on high-performance, massively parallel electronic structure theory.
 - Software products:
 - CHEMIST — <https://github.com/NWChemEx/Chemist>
 - INTEGRALS — <https://github.com/NWChemEx/Integrals>
 - PARALLELZONE — <https://github.com/NWChemEx/ParallelZone>
 - PLUGINPLAY — <https://github.com/NWChemEx/PluginPlay>
 - SCF — <https://github.com/NWChemEx/SCF>
 - SIMDE — <https://github.com/NWChemEx/SimDE>
 - TENSORWRAPPER — <https://github.com/NWChemEx/TensorWrapper>
 - WEAKLYTYPEDFLOAT — <https://github.com/NWChemEx/WeaklyTypedFloat>
- **GhostFragment** <https://github.com/rmrresearch/GhostFragment>
2021–present
 - *Founder, architect, project manager, and lead developer*
 - Software for massively parallel generalized many-body expansion and basis-set superposition corrections.
 - Contributions: concept, design, CI infrastructure, initial implementations.
- **CMakePP** <https://github.com/CMakePP>
2019–present
 - *Founder, architect, project manager, and lead developer*
 - Object-oriented CMake build system.
 - Contributions: concept, design, CI infrastructure, CMakePP language
 - Software products:
 - CMINX — <https://github.com/CMakePP/CMinx>
 - CMAKETEST — <https://github.com/CMakePP/CMakeTest>
 - CMAKEPPLANG — <https://github.com/CMakePP/CMinx>
 - CMAIZE — <https://github.com/CMakePP/CMaize>
- **Psi4** <https://psicode.org/>
2014–2017
 - *Developer*
 - Electronic structure package focused on providing a user-friendly experience.
 - Contributions: SCF, infrastructure, many-body expansion
- **Q-Chem** <https://www.q-chem.com/>
2008–2013
 - *Developer*
 - Commercial electronic structure package focused on good performance on workstations and small clusters.
 - Contributions: QM/MM

Expertise

- **Computational Chemistry Topics:**
wave function methods, fragment-based methods
- **Productivity Tools:**
Git, GitHub, VSCode, Slack
- **Computer Languages:**
C/C++, Python, CMake, L^AT_EX, Bash
- **Markup Languages:**
Doxygen, Sphinx-flavored reStructured Text, Markdown
- **Python Packages:**
Jupyter Notebooks, Matplotlib, Numpy, Sphinx
- **C++ Libraries:**
Boost, Catch2, Cereal, PyBind11
- **High-Performance Computing Libraries:**
BLAS, LAPACK, MPI

Honors and Awards

- **Better Scientific Software Fellow** **BSSw.io**
Recipient of 1 of 6 BSSw Fellowships 2024
 - Recognizes work for promoting better scientific software.
 - Awarded for: Multi-Project CI/CD for modular scientific software
- **Ames National Laboratory Scientist Leadership Development Program** **Ames, IA**
Accepted into 2022 – 2023 cohort 2022 – 2023
 - Nominees are junior level scientists at the laboratory who show leadership potential.
 - Program focuses on mentoring and developing leadership skills with a focus on career advancement at a Department of Energy National Laboratory.
- **ISU Research Collaboration Catalysts** **Ames, IA**
Accepted into 2020 – 2021 cohort 2020 – 2021
 - Nominees demonstrate potential to lead future high-impact interdisciplinary research teams.
 - Provided educational opportunities to refine leadership skills.
- **Finalist for the Emerging Technology in Computational Chemistry Symposium** **Philadelphia, PA**
252nd American Chemical Society National Meeting and Exposition 2016
- **The Ohio State University Graduate School Fellowship** **Columbus, OH**
The Ohio State University 2008–2009
- **Best Undergraduate Presentation** **South Euclid, OH**
Notre Dame College 2007
- **Cleveland State University Honors Program Scholarship** **Cleveland, OH**
Cleveland State University 2004–2008

Active Grants

- **Argonne Leadership Computing Facility Director's Discretionary Allocation** **Ames, IA**
Ryan M. Richard (*PI*) 2026
Renewal: Speeding up the computation of long-range electrostatic interactions in NWChemEx
Value of award: 60K Node Hours on Aurora
- **Ames National Laboratory Directed Research and Development FY2026-xxxx-01** **Ames, IA**
Ryan M. Richard (*PI*) 2026
Quantum Harmony: Embracing noisy quantum computing
Value of award: \$226,000.00
- **ISU LAS Frontier Science Award** **Ames, IA**
Theresa Windus (PI) and Ryan M. Richard (co-PI) 2025–2027
Error-Bound Atomistic Simulations for Reliable Machine Learning Models
Value of award: \$100,000.00
- **NSF CHE-2348724** **Ames, IA**
Theresa Windus (PI) and Ryan M. Richard (co-PI) 2024–2027
REU Site: Sustainability Institute for Machine Learning and Collaborative Open-Source Development of Enzymatic Simulations
Value of award: \$465,000.00

Completed Grants

- **Argonne Leadership Computing Facility Director's Discretionary Allocation** **Ames, IA**
Ryan M. Richard (*PI*) 2025–2026
Renewal: Speeding up the computation of long-range electrostatic interactions in NWChemEx
Value of award: 40K Node Hours on Aurora

- **Argonne Leadership Computing Facility Director's Discretionary Allocation** **Ames, IA**
2025
 Ryan M. Richard (*PI*)
 Speeding up the computation of long-range electrostatic interactions in NWChemEx
 Value of award: 100K Node Hours on Aurora
- **Ames National Laboratory Directed Research and Development FY2025-xxxx-01** **Ames, IA**
2024–2025
 Ryan M. Richard (*PI*)
 Data-Based Approach to Performance and Power Optimization
 Value of award: \$250,000.00
- **BSSw Fellowship FY2023-407** **Ames, IA**
2024–2025
 Ryan M. Richard (*PI*)
 Multi-Project CI/CD for Modular Scientific Software
 Value of award: \$50,000.00
- **Ames National Laboratory Directed Research and Development FY2024-xxxx-01** **Ames, IA**
2024
 Ryan M. Richard (*PI*)
 Automating Parameter Selection of Fragment Based Methods for Materials Challenges
 Value of award: \$287,000.00
- **Ames National Laboratory Directed Research and Development FY2022-xxxx-01** **Ames, IA**
2021–2022
 Ryan M. Richard (*PI*)
 GhostFragment: Making Strong Electron Correlation Less Scary
 Value of award: \$48,000.00
- **Renewal of XSEDE Research Allocation number CHE150006** **Atlanta, GA**
2016–2017
 Ryan M. Richard (*PI*), C. D. Sherrill (*Co-PI*).
 Accurate Crystal Lattice Energies Via Pleasantly Parallel Methods
 Value of award: \$115,006.00
- **XSEDE Research Allocation number CHE150006** **Atlanta, GA**
2015–2016
 Ryan M. Richard (*PI*), C. D. Sherrill (*Co-PI*).
 Applying Quantum Chemistry to Condensed Phase Systems Using the Many-Body Expansion
- **XSEDE Startup Allocation number CHE140145** **Atlanta, GA**
2014–2015
 Ryan M. Richard (*PI*), C. D. Sherrill (*Co-PI*).
 Accurate Crystal Properties Via Massively Parallel Quantum Chemistry

Professional Service

- **National Laboratory Information Technology (NLIT) Summit** 2024–present
Member of the Sustainable Scientific Software (S3C) organizing team.
 Helped choose topics for conference and downselect contributed presentations.
- **256th National American Chemical Society Meeting.** **Boston, MA**
2018
Session chair
- **Article peer review for:**
ACS In Focus, Journal of Chemical Education, Journal of Chemical Physics, Journal of Chemical Theory and Computation
- **Grant reviewer for:**
Better Scientific Software Fellowship, Lawrence Livermore National Laboratory Computing Grand Challenge

Ames National Laboratory Service

- **AMES Data Scientist Search Committee** **Ames National Laboratory**
2026
Committee Member
 Helped prepare interview questions and screen candidates
- **Learning Session: AI at the Lab** **Ames National Laboratory**
2024
Panelist.
 Provided a brief overview of what AI/ML can/cannot do for research and the effort needed to train an AI/ML model.

- **2022 Iowa State Fair** **Des Moines, IA**
Ames National Laboratory and Iowa State University Exhibit. 2022
Explained research at Ames National Laboratory to general public and answered questions.
- **Department of Energy's National Science Bowl** **Ames, IA**
Moderator and Judge 2019–Present
Helped setup/teardown and served as moderator and judge.

Iowa State University Service

- **Preliminary Examination Committee Member** **Ames, IA**
Krishna Suresh (substitute) 2025

Georgia Institute of Technology Service

- **PURA (President's Undergraduate Research Award) application reviewer** **Atlanta, GA**
Georgia Institute of Technology 2016
- **2nd Annual Postdoctoral Research Symposium** **Atlanta, GA**
Session chair 2015
- **1st Annual Postdoctoral Research Symposium** **Atlanta, GA**
Session chair 2014
- **'Postdocs@Tech'** **Atlanta, GA**
Co-Organizer 2014–2017
Organized monthly social events and an annual university research symposium for postdocs at Georgia Tech.

Students Mentored

Graduate Students.....

- **Anna Diaz** **Ames, IA**
Research focus: uncertainty in eigen solvers, Summer Internship 2025
- **Shalith Chamantha** **Ames, IA**
Research focus: Quantum chemistry uncertainty propagation 2024 – Present
- **Rojas, Felix** **Ames, IA**
Research focus: Numerical optimization 2024 – Present
- **Heflin, Jacob** **Ames, IA**
Research focus: Benchmarking fragment-based methods 2023 – 2026

Undergraduate Students.....

- **Ella Bushman** **Ames, IA**
Research focus: geometry optimization of small peptides 2026
- **Aspen Meissner** **Ames, IA**
Research focus: detecting the primary structure of peptides 2026
- **Daniel Woodard** **Ames, IA**
Research focus: predicting fragments of peptides. 2025
- **Rishita Kollu** **Ames, IA**
Research focus: Designing a composite method library. 2024
- **John Lewis** **Ames, IA**
Research focus: Benchmarking geometry optimizers. 2023–Present
- **Walker Hayes** **Ames, IA**
Research focus: Implementing capping methods in GhostFragment 2023

○ Issac Van Orman <i>Research focus: Implementing capping methods in GhostFragment</i>	Ames, IA 2023
○ Theodore Davis <i>Research focus: Object-oriented CMake.</i>	Ames, IA 2020–2021
○ Emin Okic <i>Research focus: Continuous integration for CMake.</i>	Ames, IA 2020–2021
○ Allison Finger <i>Research focus: Object-oriented CMake.</i>	Ames, IA 2020
○ Blake Mulnix <i>Research focus: Object-oriented CMake.</i>	Ames, IA 2020
○ Jacob Brunton <i>Research focus: Improving self-consistent field guesses.</i>	Ames, IA 2020
○ Branden Butler <i>Research focus: CMake documentation and unit testing</i>	Ames, IA 2019–Present
○ Andres Garcia-Alejo <i>Research focus: Machine learning molecular structure</i>	Ames, IA 2018–2019
○ Brodie Schroeder <i>Research focus: Machine learning applications to quantum chemistry</i>	Ames, IA 2017
○ Michael Zott <i>Research focus: Understanding the many-body expansion</i>	Atlanta, GA 2015–2017

Other Mentoring Activities

○ Anna Diaz <i>Graduate Student Summer Internship. Research focus: uncertainty in eigen solvers</i>	Ames, IA 2025
○ Nancy L. Glenn Griesinger <i>Visiting Faculty Program. Research focus: uncertainty in electron repulsion integrals.</i>	Ames, IA 2025

Invited Presentations

2025.....

19. **Ryan M. Richard.** *TensorWrapper Wishful Thinking?*. Toulouse Tensor Workshop. Toulouse, France. August 2025
18. **Ryan M. Richard.** *NWChemEx: Towards a fully reusable electronic structure package.* Majvik, Finland. July 2025

2024.....

17. **Ryan M. Richard.** *The NWChemEx Community's contributions to performance portable, modular, reusable, and interoperable computational chemistry.* The 75th Southeastern Regional Meeting of the American Chemical Society. Atlanta, GA. October 2024.
16. **Ryan M. Richard.** *PluginPlay: Enabling high- performance scientific software one module at a time.* CECAM Flagship Workshop. Lausanne, Switzerland. February 2024.

2023.....

15. **Ryan M. Richard.** *NWChemEx: Designing a Computational Chemistry App Store for The Exascale.* APS March Meeting. Las Vegas, NV. March 2023.

14. **Ryan M. Richard.** *NWChemEx: Challenges Faced in Designing an Electronic Structure Program for the Exascale.* SIAM Conference on Computational Science and Engineering. Amsterdam, New Holland. The Netherlands. February 2023.
2022.....
13. **Ryan M. Richard.** *Software design for the exascale era: The NWChemEx perspective.* 2022 Midwest Regional Meeting for the American Chemical Society. October 2022.
2019.....
12. **Ryan M. Richard.** *The SDE: A General Computational Chemistry Software Framework.* Blue Waters Webinar. March 2019.
11. **Ryan M. Richard** and Theresa L. Windus. *CMake Packaging Project: Reliable, Reproducible, and Reusable Build Systems Made Easy.* Spring NWCHEMEX Team Meeting. Richland, WA. March 2019.
10. **Ryan M. Richard.** *The Simulation Development Environment (SDE): A C++ Framework for Reusable Computational Chemistry.* Society for Industrial and Applied Mathematics Conference on Computational Science and Engineering. Spokane, WA. February 2019.
2018.....
9. **Ryan M. Richard.** *Applying Ab Initio Methods to Large Molecules.* Ames Laboratory. Ames, IA. August 2018.
2017.....
8. **Ryan M. Richard.** *Exascale Prototyping Via A Computational Chemistry App Store.* 254th National American Chemical Society Meeting. Washington, D.C. August 2017.
2016.....
7. **Ryan M. Richard.** *Pulsar: A Computational Chemistry App Store.* PS14 Developers Meeting. University of Georgia. Athens, GA. November 2016.
6. **Ryan M. Richard.** *ForcemanII: Status and Possible Future Directions.* Q-CHEM Developers Meeting. University of Pennsylvania. Philadelphia, PA. August 2016.
5. **Ryan M. Richard.** *Understanding Chemistry Via Fragment Based Methods.* University of Tennessee. Knoxville, TN. February 2016.
4. **Ryan M. Richard.** *Understanding Chemistry Via Fragment Based Methods.* Youngstown State University. Youngstown, OH. February 2016.
3. **Ryan M. Richard.** *Understanding Chemistry Via Fragment Based Methods.* Tennessee Technological University. Cookeville, TN. January 2016.
2015.....
2. **Ryan M. Richard.** *Understanding Chemistry Via Fragment Based Methods.* University of North Florida. Jacksonville, FL. November 2015.
2008.....
1. **Ryan M. Richard, David W. Ball.** *Thermodynamic Studies on the Potential Use of Boron- and/or Nitrogen-Containing Molecules as New High Energy Materials.* Ohio State University. Columbus, OH. February 2008.

Contributed Presentations

2025.....

22. **Ryan M. Richard**. *GhostFragment: a reusable, flexible, modular library for generalized fragment-based methods*. Pacificchem 2025. Honolulu, HI. December 2025.

2019.....

21. **Ryan M. Richard** and Theresa L. Windus. *CMake Packaging Project: Reliable, Reproducible, and Reusable Build Systems Made Easy*. 257th National American Chemical Society Meeting. Orlando, FL. April 2019.
20. **Ryan M. Richard**, Kristopher Keipert, Thom Dunning Jr., Robert Harrison, and Theresa L. Windus. *The NWChemEx Simulation Development Environment - A General Computational Chemistry Framework*. Poster Presentation. Society for Industrial and Applied Mathematics Conference on Computational Science and Engineering. Spokane, WA. February 2019.

2018.....

19. **Ryan M. Richard** and Theresa L. Windus. *Leveraging NWChemEx's computational chemistry app store to design an exascale SCF*. Oral Presentation. 256th National American Chemical Society Meeting. Boston, MA. August 2018.
18. **Ryan M. Richard**, Brandon W. Bakr, and C. David Sherrill. *Understanding the Many-Body Basis Set Superposition Error: Beyond Boys and Bernardi*. Oral Presentation. 50th Midwest Theoretical Chemistry Conference. Chicago, IL. June 2018.

2017.....

17. **Ryan M. Richard**. *Leveraging a Computational Chemistry App-Store for Both Teaching and Researching Chemistry*. Oral Presentation. 254th National American Chemical Society Meeting. Washington, D.C. August 2017.

2016.....

16. **Ryan M. Richard**, Ben Pritchard, and C. D. Sherrill. *Leveraging a Computational Chemistry App Store to Compute High Accuracy Lattice Energies of Molecular Crystals*. Oral Presentation. 252nd American Chemical Society National Meeting and Exposition. Philadelphia, PA. August 2016
15. **Ryan M. Richard**, Brandon Bakr, and C. D. Sherrill. *Understanding Basis Set Superposition Error in Many-Body Systems: Beyond Boys and Bernardi*. Oral Presentation. 47th Southeast Theoretical Chemistry Association Annual Meeting. Tallahassee, FL. May 2016

2015.....

14. **Ryan M. Richard**, Michael S. Marshall, Jean-Luc Brédas, Noa Marom, and C. D. Sherrill. *CCSD(T)/CBS Benchmark Quality Ionization Potentials (IP) and Electron Affinities (EA)*. Poster Presentation. 2nd Annual Postdoctoral Research Symposium. Atlanta, GA. October 2015.
13. **Ryan M. Richard**, C. D. Sherrill. *Massively Parallel Fragment Based Methods as Implemented in PSI4*. Oral Presentation. 250th American Chemical Society National Meeting and Exposition. Boston, Massachusetts. August 2015.
12. **Ryan M. Richard**, Michael S. Marshall, Jean-Luc Brédas, Noa Marom, and C. D. Sherrill. *CCSD(T)/CBS Benchmark Quality Ionization Potentials (IP) and Electron Affinities (EA)*. Poster Presentation. 46th Southeast Theoretical Chemistry Association Annual Meeting. Orlando, Florida. May 2015.

2014.....

11. **Ryan M. Richard**, C. D. Sherrill. *Achieving Chemical Understanding Via High Performance Computing*. Oral Presentation. 1st Annual Postdoctoral Research Symposium. Atlanta, Georgia. September 2014.

2013.....

10. **Ryan M. Richard**, J. M. Herbert. *Achieving the Complete-Basis Limit in Large Molecular Clusters: Computationally Efficient Procedures to Eliminate Basis-Set Superposition Error*. Oral Presentation. 246th American Chemical Society National Meeting and Exposition. Indianapolis, Indiana. September 2013.
9. **Ryan M. Richard**, J. M. Herbert. *Achieving the Complete-Basis Limit in Large Molecular Clusters: Computationally Efficient Procedures to Eliminate Basis-Set Superposition Error*. Oral Presentation. 68th International Symposium on Molecular Spectroscopy. Columbus, Ohio. June 2013.
8. **Ryan M. Richard**, J. M. Herbert. Oral Presentation. 45th Midwest Theoretical Chemistry Conference. Champaign, Illinois. May 2013.

2012.....

7. **Ryan M. Richard**, J. M. Herbert. *A Generalized Many-Body Expansion and a Unified View of Fragment-Based Methods in Electronic Structure Theory*. Poster presentation. 44th Midwest Theoretical Chemistry Conference. Madison, Wisconsin. June 2012.

2011.....

6. **Ryan M. Richard**, J. M. Herbert. *TDDFT Calculations of Transient IR Spectra of DNA*. Oral Presentation. 66th International Symposium on Molecular Spectroscopy. Columbus, Ohio. June 2011.
5. **Ryan M. Richard**, J. M. Herbert. *Time-Dependent Density-Functional Description of the 1L_a State in Polycyclic Aromatic Hydrocarbons: Charge-Transfer State in Disguise?*. Poster Presentation. 43rd Midwest Theoretical Chemistry Conference. South Bend, Indiana. June 2011.

2010.....

4. **Ryan M. Richard**, J. M. Herbert. *Determination of Exciton Length in Aqueous B-DNA Using Long-Range Time-Dependent Density Functional Theory (LRC-TDDFT)*. Oral Presentation. 65th International Symposium on Molecular Spectroscopy. Columbus, Ohio, June 2010.

2008.....

3. **Ryan M. Richard**, D. W. Ball. *Thermodynamic Studies on Boron and Nitrogen Containing Spiropentanes*. Oral Presentation. American Chemical Society Meeting in Miniature. Oberlin College. Oberlin, OH. March 2008.

2007.....

2. **Ryan M. Richard**, D. W. Ball. *G2, G3, and Complete Basis Set Calculations on Small Hydroboranes and Mixed Boron-Nitrogen Containing Rings*. American Chemical Society Meeting in Miniature. Notre Dame College. South Euclid, OH. March 2007. Winner of Best Undergraduate Presentation.
1. **Ryan M. Richard**, D. W. Ball. *New Potential High Energy Materials*. Cleveland State University Undergraduate Research Symposium. Summer 2007

Miscellaneous Workshops and Presentations

2. Organizer for Virtual Workshop on Multi-Project CI/CD. Virtual. 2024.
1. **Ryan M. Richard**. *Best Practices for Multi-project CI/CD*. NLIT (National Laboratory Information Technology) Summit 2024. Seattle, Washington. April 2024.

Non Peer-Reviewed Publications

3. **Ryan M. Richard**. *Best Practices for Multi-Project Continuous Integration and Deployment*. https://bssw.io/blog_posts/best-practices-for-multi-project-continuous-integration-and-deployment.
2. **Ryan M. Richard**, T. L. Windus. *Is a Language Barrier Impeding Development of Better Scientific Software?*. ASCR (Advanced Scientific Computing Research) Workshop on the Science of Scientific-Software Development and Use. Virtual. December 13-15 2021.
1. **Ryan M. Richard**, Z. Crandall, H. v. Dam, N. Govind, K. Kowalski, T. L. Windus. *Stewardship of Software for Scientific and High-Performance Computing*. Office of Advanced Scientific Computing Research.

Review Articles

2. **Ryan M. Richard**, K. U. Lao, J. M. Herbert. *Aiming for Benchmark Accuracy with the Many-Body Expansion*. *Accounts of Chemical Research*. 47 (2014) 2828-2836.
1. L. D. Jacobson, **Ryan M. Richard**, K. U. Lao, J. M. Herbert. *Efficient Monomer-Based Quantum Chemistry Methods for Molecular and Ionic Clusters*. *Annual Reports of Computational Chemistry*. 9 (2013) 25-56.

Publications

2025.....

40. K. Fujioka, **Ryan M. Richard**, J. Waldrop, Y. Lou, T. L. Windus, and R. Sun. *VENUSpy: A Chemical Dynamics Simulation Program in the Era of Machine Learning and Exascale Computing*. *Journal of Chemical Theory and Computation*. 21 (2025) 10679.
39. J. M. Waldrop, W. de Jong, E. F. Valeev, T. L. Windus, and **Ryan M. Richard**. *Chemist: A Domain-Specific Language by Chemists for Chemists*. *The Journal of Physical Chemistry A*. 129 (2025) 7964.
38. C. Stevens, K. Kjeer, **Ryan M. Richard**, E. Valeev, M. Cohen. *Model Assisted Refinement of Metamorphic Relations for Scientific Software*. ICSE 2025 New Ideas and Emerging Results (Conference Proceeding).
37. J. M. Waldrop, **Ryan M. Richard**. *Sigma: Uncertainty Propagation for C++*. *Journal of Open Source Software*. 10 (2025) 7404.

2024.....

36. Z. Crandall, T. L. Windus, **Ryan M. Richard**. *CMake: Simplifying inter-package modularity from the build up*. *Journal of Chemical Physics*. 160 (2024) 092502.

2023.....

35. V. Gavini ... **Ryan M Richard** ... D. Perez. *Roadmap on electronic structure codes in the exascale era*. *Modelling and Simulation in Materials Science and Engineering*. 31 (2023) 063301. (30th author of 41)
34. **Ryan M. Richard**, K. Keipert, J. Waldrop, M. Keçeli, D. Williams-Young, R. Bair, J. Boschen, Zachery Crandall, Kevin Gasperich, Quazi Ishtiaque Mahmud, A. Panyala, E. Valeev, H. van Dam, W A. de Jong, T. L. Windus. *PluginPlay: Enabling exascale scientific software one module at a time*. *The Journal of Chemical Physics*. 158 (2023) 184801.

2022.....

33. B. Butler and **Ryan M. Richard**. *CMinx: A CMake Documentation Generator*. *Journal of Open Source Software*. 7 (2022) 4680.

2021.....

32. E. Epifanovsky, ..., **Ryan M. Richard**, ..., A. I. Krylov. *Software for the frontiers of quantum chemistry: An overview of developments in the Q-Chem 5 package*. The Journal of Chemical Physics. 155 (2021) 084801. (139th of 220 authors).
31. Karol Kowalski, ..., **Ryan M. Richard**, ..., Theresa L. Windus. *From NWChem to NWChemEx: Evolving with the Computational Chemistry Landscape*. Chemical Reviews. 121 (2021) 4962- 4998. (19th of 29 authors).

2020.....

30. Edoardo Apra, ..., **Ryan Richard**, ..., and Robert Harrison . *NWChem: Past, Present, and Future*. Journal of Chemical Physics. 152 (2020) 184102. (82nd of 114 authors).

2019.....

29. **Ryan M. Richard**, Colleen Bertoni, Jeffery S. Boschen, Kristopher Keipert, Benjamin Pritchard, Edward F. Valeev, Robert J. Harrison, Wibe A. de Jong, and Theresa L. Windus. *Developing a Computational Chemistry App Store for the Exascale Era*. Computing in Science & Engineering. 21 (2019) 48.

2018.....

28. **Ryan M. Richard**, Brandon Bakr, and C. David Sherrill. *Understanding the Many-Body Basis Superposition Error: Beyond Boys and Bernardi*. Journal of Chemical Theory and Computation. 14 (2018) 2386.

2017.....

27. Robert M. Parrish, ..., **Ryan M. Richard**, ... *Psi4 1.1: An Open-Source Electronic Structure Program Emphasizing Automation, Advanced Libraries, and Interoperability*. Journal of Chemical Theory and Computation. 13 (2017) 3185. (10th of 26 authors)

2016.....

26. Ka Un Lao , Kuan-Yu Liu , **Ryan M. Richard** , and John M. Herbert. *Understanding the many-body expansion for large systems. II. Accuracy considerations*. Journal of Chemical Physics. 144 (2016) 164105.
25. O. Dolgounitcheva, Manuel Díaz-Tinoco, V. G. Zakrzewski, **Ryan M. Richard**, Noa Marom, C. David Sherrill and J. V. Ortiz. *Accurate Ionization Potentials and Electron Affinities of Acceptor Molecules IV: Electron-Propagator Methods*. Journal of Chemical Theory and Computation. 12 (2016) 627-637.
24. **Ryan M. Richard**, Michael S. Marshall, Olga Dolgounitcheva, J. V. Ortiz, Jean-Luc Brédas, Noa Marom, and C. David Sherrill. *Accurate Ionization Potentials and Electron Affinities of Acceptor Molecules. I. Reference Data at the CCSD(T) Complete Basis Set Limit*. Journal of Chemical Theory and Computation. 12 (2016) 595-604.

2015.....

23. Y. Shao, ..., **Ryan M. Richard**,... *Advances in Molecular Quantum Chemistry Contained in the Q-Chem 4 Program Package*. Molecular Physics. 113 (2015) 184-215. (22nd author out of 150).

2014.....

22. **Ryan M. Richard**, K. U. Lao, J. M. Herbert. *Understanding the Many-Body Expansion for Large Systems. I. Precision Considerations*. The Journal of Chemical Physics. 141 (2014) 014108:1-14.

2013.....

21. Z. C. Holden, **Ryan M. Richard**, J. M. Herbert. *Periodic Boundary Conditions for QM/MM Calculations: Ewald Summation for Extended Gaussian Basis Sets*. The Journal of Chemical Physics. 139 (2013) 244108:1-13.

20. **Ryan M. Richard**, K. U. Lao, J. M. Herbert. *Approaching the Complete-Basis Limit with a Truncated Many-Body Expansion*. The Journal of Chemical Physics. 139 (2013) 224102:1-11.
 19. **Ryan M. Richard**, K. U. Lao, J. M. Herbert. *Achieving the CCSD(T) Basis-Set Limit in Sizable Molecular Clusters: Counterpoise Corrections for the Many-Body Expansion*. The Journal of Physical Chemistry Letters. 4 (2013) 2674-2680.
 18. **Ryan M. Richard**, J. M. Herbert. *Many-Body Expansion with Overlapping Fragments: Analysis of Two Approaches*. Journal of Chemical Theory and Computation. 9 (2013) 1408-1416.
- 2012.....
17. **Ryan M. Richard**, J. M. Herbert. *A Generalized Many-Body Expansion and a Unified View of Fragment-Based Methods in Electronic Structure Theory*. The Journal of Chemical Physics. 137 (2012) 064113.
 16. W. Morales, K. W. Street Jr., **Ryan M. Richard**, D. J. Valco. *Tribological Testing and Thermal Analysis of an Alkyl Sulfate Series of Ionic Liquids for Use as Aerospace Lubricants*. Tribology Transactions. 55 (2012) 815-821.
- 2011.....
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